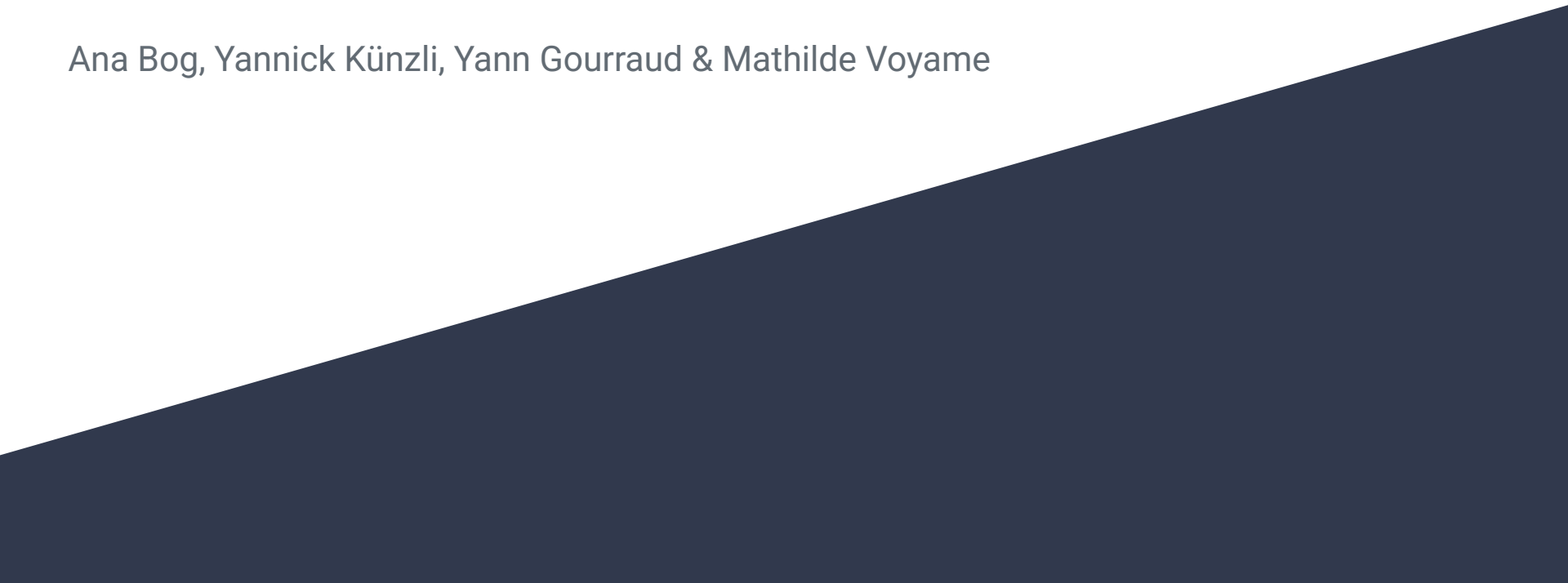


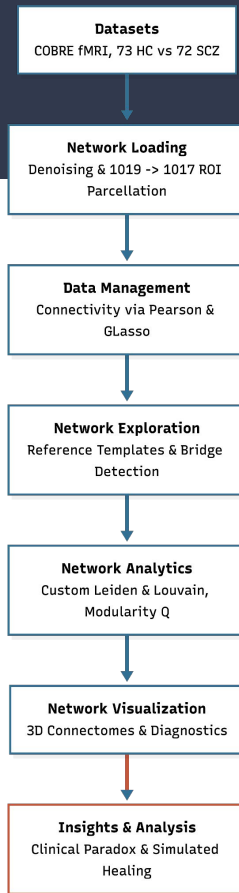
Social Media Analytics Schizo-Nets

Ana Bog, Yannick Künzli, Yann Gourraud & Mathilde Voyame

A dark blue diagonal gradient bar that starts from the bottom left corner and extends towards the top right corner, covering the lower half of the slide.

Introduction

- Analyze schizophrenia using graph theory :
node = brain region, edge = connectivity
between regions
- Assumption : schizophrenia affects the
modular organization of the brain
- “Healing” : identify abnormal bridges in
schizophrenic patients and remove them in a
simulation



Cobre Dataset

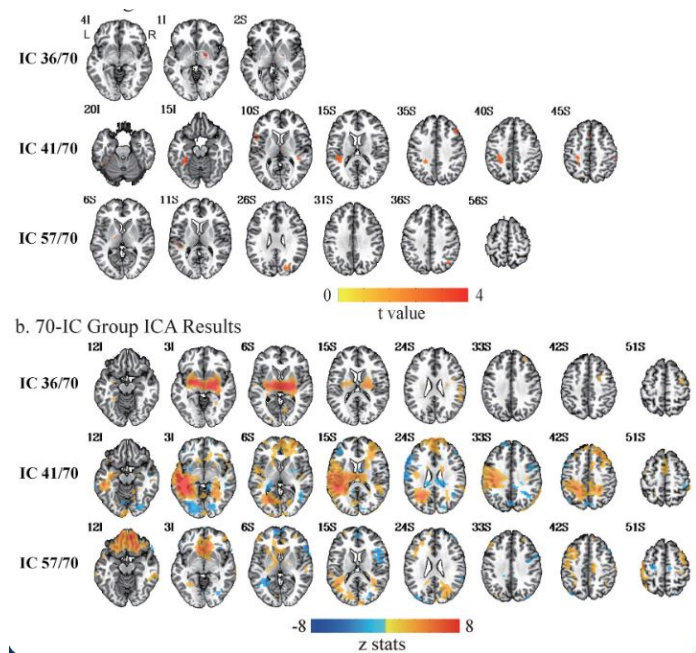
4D Structure (Voxel x time)

Already preprocessed via NIAK (a dedicated pipeline designed for fMRI).

Dimensions: 3D Voxels (Space) $\times$ Time (Time series).

Represented as a spatio-temporal 4D Tensor.

Cohort: 145 subjects (72 SCZ patients / 73 HC controls).



Network Loading

Atlas Fusion & ROI Definition

Combined 1000 Schaefer parcels + 19 Harvard-Oxford subcortical regions → 1017 total ROIs.

Signal Aggregation

Calculated the mean intensity across all constituent voxels for each brain region.

Connectivity Estimation

Assessed functional coupling via time series correlation (Pearson vS GLasso) across all regional nodes.

Numerical Stability (Dead Columns)

Handled signal drop-outs with Gaussian Noise injection to prevent convergence failure in downstream algorithms.

Database Persistence

Streamlined storage of 4D data into a unified HDF5 Connectome Database.

Connectivity: Pearson vs. GLasso

- **Pearson Correlation**
- **Marginal Association**
 - Captures **all types of coactivity**, including indirect paths.
 - Results in **Hyper-Dense** networks.
- **Graphical Lasso**
 - Scikit-learn GraphicalLassoCV
- **Conditional Association**
 - Isolates **partial connections**, driving indirect links to zero.
 - Enforces **parsimonious sparsity**.

Network Analytics (1), Algorithms

Feature	Louvain	Leiden
Phases per pass	2 Phases: - Local Node Moving - Graph Aggregation	3 Phases: - Local Node Moving - Partition Refinement - Graph Aggregation
Community Structure	Can produce internally disconnected communities.	Guaranteed to produce strongly connected communities.

Network Analytics (2), Metrics

Dataset	Algorithm	Modularity	NMI	Jaccard	Runtime (s)	Communities	Within Corr	Between Corr	W/B Ratio
hc_glasso	Custom Leiden	0.709	0.821	0.581	0.392	45.2	0.0074	0.0002	45.1
hc_glasso	Custom Louvain	0.710	0.806	0.584	0.161	45.2	0.0073	0.0002	45.5
hc_glasso	Library Leiden	0.721	0.825	0.612	0.030	47.2	0.0078	0.0002	49.2
hc_glasso	NX Louvain	0.713	1.000	1.000	0.045	45.4	0.0075	0.0002	46.5
hc_pearson	Custom Leiden	0.139	0.771	0.738	70.360	4.2	0.3363	0.1795	1.9
hc_pearson	Custom Louvain	0.139	0.712	0.679	34.338	3.6	0.3235	0.1788	1.9
hc_pearson	Library Leiden	0.140	0.825	0.817	0.982	4.0	0.3336	0.1794	1.9
hc_pearson	NX Louvain	0.139	1.000	1.000	1.834	4.2	0.3334	0.1814	1.9
scz_glasso	Custom Leiden	0.750	0.881	0.641	0.514	280.8	0.0070	0.0001	2422.5
scz_glasso	Custom Louvain	0.751	0.878	0.659	0.088	281.8	0.0072	0.0001	2055.1
scz_glasso	Library Leiden	0.757	0.888	0.688	0.022	281.8	0.0072	0.0001	2133.2
scz_glasso	NX Louvain	0.753	1.000	1.000	0.031	281.6	0.0071	0.0001	2127.5
scz_pearson	Custom Leiden	0.120	0.795	0.776	69.456	4.0	0.3139	0.1818	1.8
scz_pearson	Custom Louvain	0.120	0.824	0.790	36.055	3.6	0.3023	0.1831	1.7
scz_pearson	Library Leiden	0.121	0.869	0.858	0.902	4.0	0.3141	0.1826	1.8
scz_pearson	NX Louvain	0.120	1.000	1.000	1.979	4.0	0.3118	0.1830	1.8

- Modularity (Q): Measures how clearly the brain divides itself into distinct, isolated communities
- NMI & Jaccard: Algorithm accuracy, how closely the partition matches the baseline.
- Within/between ratio: A high ratio shows the detected communities are mathematically distinct from one another
- Runtime
- Number of communities

Network Analytics (3), Results

Dataset	Algorithm	Modularity	NMI	Jaccard	Runtime (s)	Communities	Within Corr	Between Corr	W/B Ratio
hc_glasso	Custom Leiden	0.709	0.821	0.581	0.392	45.2	0.0074	0.0002	45.1
hc_glasso	Custom Louvain	0.710	0.806	0.584	0.161	45.2	0.0073	0.0002	45.5
hc_glasso	Library Leiden	0.721	0.825	0.612	0.030	47.2	0.0078	0.0002	49.2
hc_glasso	NX Louvain	0.713	1.000	1.000	0.045	45.4	0.0075	0.0002	46.5
hc_pearson	Custom Leiden	0.139	0.771	0.738	70.360	4.2	0.3363	0.1795	1.9
hc_pearson	Custom Louvain	0.139	0.712	0.679	34.338	3.6	0.3235	0.1788	1.9
hc_pearson	Library Leiden	0.140	0.825	0.817	0.982	4.0	0.3336	0.1794	1.9
hc_pearson	NX Louvain	0.139	1.000	1.000	1.834	4.2	0.3334	0.1814	1.9
scz_glasso	Custom Leiden	0.750	0.881	0.641	0.514	280.8	0.0070	0.0001	2422.5
scz_glasso	Custom Louvain	0.751	0.878	0.659	0.088	281.8	0.0072	0.0001	2055.1
scz_glasso	Library Leiden	0.757	0.888	0.688	0.022	281.8	0.0072	0.0001	2133.2
scz_glasso	NX Louvain	0.753	1.000	1.000	0.031	281.6	0.0071	0.0001	2127.5
scz_pearson	Custom Leiden	0.120	0.795	0.776	69.456	4.0	0.3139	0.1818	1.8
scz_pearson	Custom Louvain	0.120	0.824	0.790	36.055	3.6	0.3023	0.1831	1.7
scz_pearson	Library Leiden	0.121	0.869	0.858	0.902	4.0	0.3141	0.1826	1.8
scz_pearson	NX Louvain	0.120	1.000	1.000	1.979	4.0	0.3118	0.1830	1.8

- Modularity: The library Leiden outperform both louvains and our custom Leiden
- NMI: Generally, Leiden finds structurally similar communities relative to the baseline
- Jaccard: Shows consistency with NMI, strong agreement between Leiden and library implementations
- Runtime: The libraries offer a significant speed improvement over our custom implementations
- The number of communities is consistent with the libraries
- W/B Ratio: In scz_glasso, Leiden achieves a significantly higher ratio. Possibly showing Leiden grouped brain regions successfully

Network Exploration –

Do SCZ and HC Brain Networks Differ? Q, ARI & NMI

- **p-value:** [0;1]; < 0.05 means the result is statistically significant
- **Cohen's d:** measures effect size in SD units; |d| $\approx$ 0.2 small, 0.5 medium, 0.8 large

A) Community Structure

1. Compute Q score on mean HC matrix using Louvain / Leiden
2. Compare Q score of mean HC matrix to that each individual SCZ patient

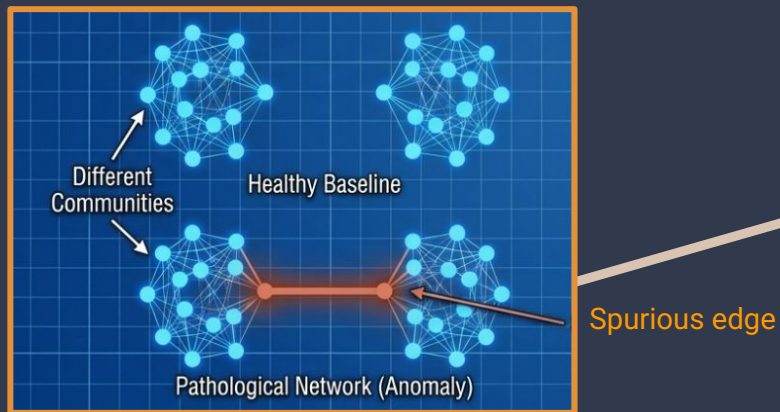
Metric	Method	HC mean	SCZ mean	Δ	p	Cohen d
Q	Glasso	0.7361	0.7452	+0.0091	0.2865	-0.089
Q	Pearson	0.2835	0.2900	+0.0066	0.5866	-0.089

B) Community Organisation

3. Build a healthy reference partition to which we would compare individual SCZ data.
4. Compute ARI & NMI to measure whether the ROIs are grouped in the right way.

ARI	Glasso	0.2687	0.2277	-0.0410	0.0138	+0.397
ARI	Pearson	0.2406	0.1857	-0.0549	0.0105	+0.456
NMI	Glasso	0.5198	0.4929	-0.0269	0.0004	+0.553
NMI	Pearson	0.2521	0.1995	-0.0526	0.0037	+0.500

Network Healing



C) Find Spurious Edges

IF:

1. 2 ROIs belonging to different communities in the healthy reference
2. Z-score > 2.0

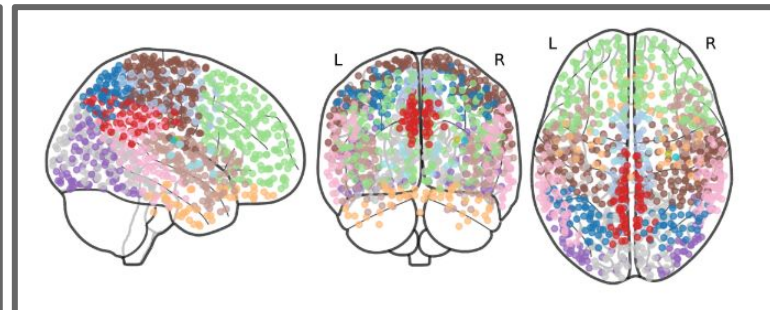
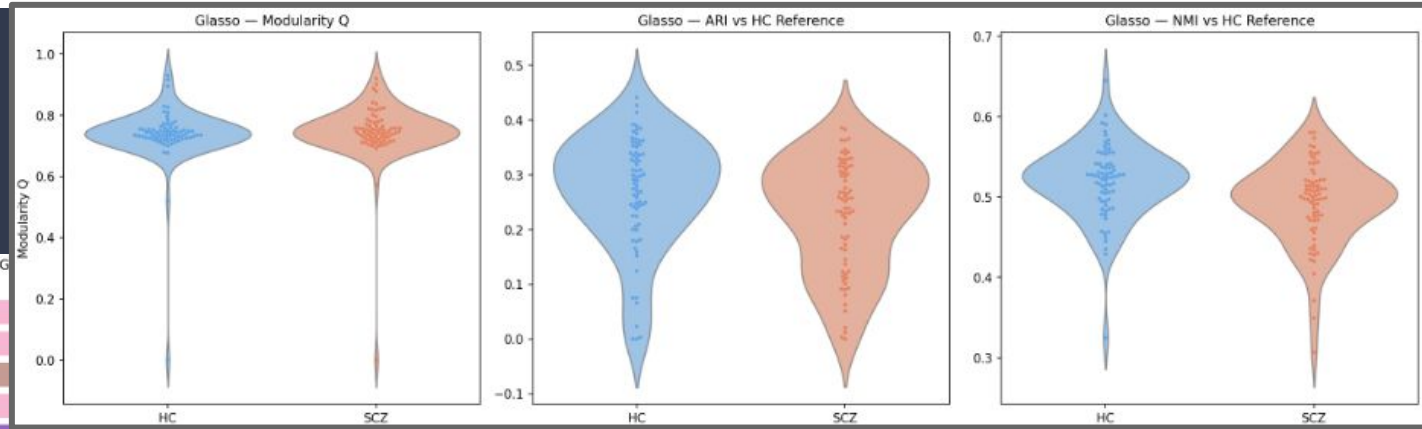
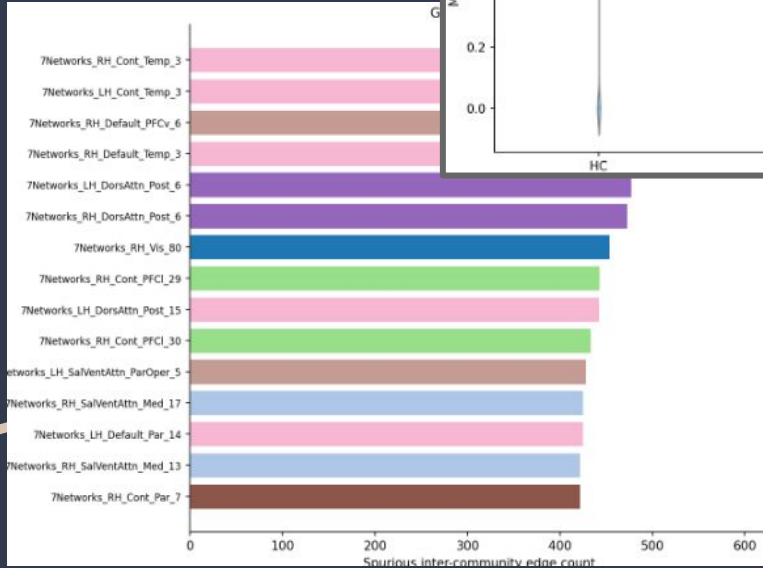
THEN: spurious edge → Rank ROIs by spurious-edge count

D) Simulated Healing

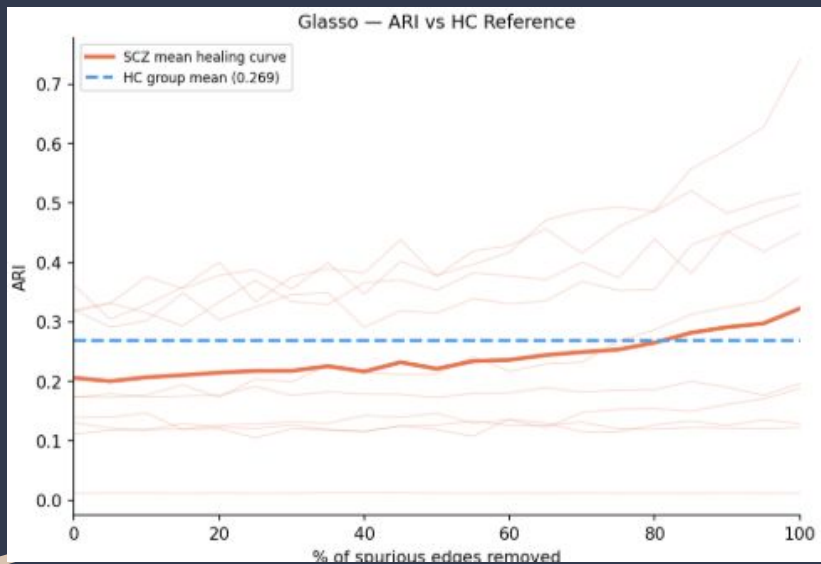
1. Delete top-k% spurious edges
2. Re-run Louvain / Leiden
3. Measure ARI

→ 21 steps, from 0% to 100%

Network Visualization



Insights & Analysis



→ DEMO

Questions ?

